

JUAN D. LUQUE CHANG

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<https://jdluque.github.io>

Education

- **PhD, Computer Science** GPA: 3.9/4.0
University of Maryland, College Park (UMD) Expected May 2025
- **BS, Mathematics & Computer Science minor** GPA 3.9/4.0
University of Maryland Baltimore County (UMBC); Sherman Scholar May 2019

Programming competencies

Core: Python, Typescript, C++, Java, MySQL, Rust

Libraries: pytorch, torch_geometric, pandas, AWS Amplify, scikit-learn, gurobi, matplotlib seaborn, NetworkX

Relevant experience

- **Data Science for Social Good (DSSG) at Carnegie Mellon University** Pittsburgh, PA
DSSG Fellow – repository: https://github.com/dssg/dojo_mh_public May 2022 - August 2022
 - Implemented a scalable *Machine Learning* pipeline, pooling data from > 100 public services heterogeneous tables, using Python, SQL, sci-kit learn and bash to predict behavioral health crises in Kansas
 - Worked closely with stakeholders (outreach teams, mental health center, hospitals, EMS, etc.) to understand the problem and piece together an appropriate solution
 - Presented a 3 minute talk at DataFest 2022
- **UMD Computer Science Department** College Park, MD
Research and Teaching Assistant August 2021 - Present
 - **Publications:** Barter Exchange with Shared Item Valuations (The Web Conference '24); Group Fairness in Set Packing (IJCAI-23); Motif Counting for Graph Neural Networks; Concentration of Submodular Functions Under Negative Dependence
 - **TA:** Machine Learning, Discrete Structures (x3), Intro to Programming Languages I and II

Projects

- **Obsidian Plugin Development (written in Typescript)** Created 2021
<https://obsidian.md> is a free markdown-based note making and editing software
 - Next Link: implements re-bindable shortcuts for jumping the cursor back and forth between note hyperlinks.
 - Share Connected Components: avoid dead links by automating the sharing of connected components of notes
- **Machine Learning Projects on MNIST, CIFAR10, and other datasets** Completed December 2020
 - Implemented DNNs, CNNs, transfer learning (on VGG16), PCA, MDA, spectral clustering, kSVM, and more

Other skills and experience

Other experiences: Google Computer Science Research Mentorship, Mentee 2021; Learning Assistant, Multivariate Calculus at Montgomery College 2017; Learning Assistant, Physics I, UMBC 2018-2019

Other competencies: Android Studio, L^AT_EX, MATLAB, Github Projects, Trello, Linux (Ubuntu, Debian), Flutter, GraphQL

Coursework: Advanced Machine Learning, Algorithmic Lower Bounds, Algorithm Design and Analysis, Advanced Nonlinear Optimization, Quantum Information Processing, Real Analysis, Complex Analysis, Measure Theory, Computational Methods, Computational Geometry

Other skills/hobbies: English (native), Spanish (native), French (elementary); MC Board Hoarders (Vice-president); 1000 lb club; bouldering

Volunteering: BlairHacks 2022 Python Workshop Host, Baltimore Polytechnic Institute Volunteer Teaching Assistant, Lakeland Elementary and Middle School Volunteer Math Coach and Teaching Assistant, George B. Thomas Learning Academy Volunteer Tutor

Awards: Freeman A. Hrabowski President's Scholarship; Sherman Scholar Scholarship; Kenneth-Wantling Endowed Scholarship; Koski-Finnerty Math/Sciences Scholarship; General Honors Scholarship; Transfer Student Alliance